Zahran Toric Tool

USER MANUAL

VERSION	PLATFORM	DATE
1.0.0	iOS / Android	May 2026

A clinical-grade mobile application for toric intraocular lens planning, IOL alignment, and astigmatism outcome analysis — designed for ophthalmic surgeons.

Table of Contents

11 Clinical Reference

12 Troubleshooting

CHAPTER 1

Introduction & Overview

What the Zahran Toric Tool does and how to use it safely.

1.1 Purpose of the Application

The **Zahran Toric Tool** is a mobile clinical decision-support tool designed to assist ophthalmic surgeons in planning and executing toric intraocular lens (IOL) implantation. It combines pre-operative biometric calculation, intraoperative axis alignment, and post-operative outcome analysis in a single cohesive workflow.

The application follows established ophthalmic conventions including TABO axis notation (0°–180°) and the double-angle vector method (DAVM) for all astigmatism calculations.

1.2 Key Features

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Toric IOL Calculator

Calculates residual astigmatism, IOL cylinder power, and optimal target axis from corneal biometry.

📷

Premium Camera Capture

Instrument-gauge dial with 310° TABO markings, live tilt sensor, and center level readout.

🎯

IOL Alignment Overlay

Real-time reference and target axis lines overlaid on the captured eye photo.

📌

Landmark Annotation

Tap-to-place vessel reference markers on the eye photo for pre-operative marking.

📊

Alpins Vector Analysis

Full post-operative outcome analysis: TIA, SIA, DV, CI, ME, AE, and IS metrics with double-angle plot.

📈

SIA Nomogram Tracker

Builds a personal SIA nomogram from your cases to personalise future calculations.

🔬

PCA Correction

Barrett-style Posterior Corneal Astigmatism correction for more accurate toric planning.

🩺

Post-Refractive Mode

Adapted planning for eyes with prior LASIK, PRK, or RK surgery.

1.3 Medical Disclaimer

⚠ Important Medical Notice

This application is intended as a **clinical decision-support tool only**. It does not replace the professional judgment of a qualified ophthalmic surgeon. All calculations must be reviewed and verified by the treating surgeon before any clinical action is taken. The developer assumes no liability for outcomes resulting from use of this application. Always cross-reference results with your IOL manufacturer's nomogram and current clinical guidelines.

CHAPTER 2

Getting Started

Installation, permissions, and your first case.

2.1 Installation & Permissions

The app requires the following device permissions:

PERMISSION	PURPOSE	REQUIRED?
Camera	Capture anterior segment eye photos for alignment	REQUIRED
Photo Library	Save captured eye images to device gallery (optional)	OPTIONAL
Motion Sensors	Device tilt detection for leveling the camera	REQUIRED

You will be prompted to grant Camera access the first time you navigate to the Camera screen. Motion sensor access is granted automatically.

2.2 Navigation Overview

The app uses a stack-based navigation system. The main entry point is the **Patient List**. From there, all other screens are accessible:

SCREEN HIERARCHY

- **Patient List** → Patient Detail → Toric Calculator → (Camera → Alignment)
- **Patient Detail** → Alpins Analysis
- **Patient Detail** → Landmark Annotation
- **Patient List** (header) → SIA Nomogram
- **Patient List** (header) → Settings

2.3 Workflow Summary

1. **Create a patient record** — enter name, DOB, and any notes.

2. **Add an eye record** — enter keratometry (K1/K2), IOL power, and cylinder data.
3. **Run the Toric Calculator** — review recommended IOL cylinder and axis.
4. **Capture the eye photo** — use the premium camera gauge to align and capture.
5. **Align the IOL** — use the alignment overlay in the operating room.
6. **Post-operatively** — enter actual refraction and run Alpins Analysis.
7. **Log the case to your SIA Nomogram** — build your personalised tracker over time.

CHAPTER 3

Patient Management

Creating and organising your patient database.

3.1 Patient List Screen

The Patient List is the home screen of the app. All patients are stored locally on your device using encrypted storage. The list shows name, date of birth, and the number of eye records.

- Tap **+ New Patient** to add a record.
- Tap any patient row to open their detail page.
- Tap the **SIA** button (top-right) to open the SIA Nomogram.
- Tap the ⚙ gear icon to open Settings.

3.2 Adding / Editing a Patient

The Patient Form captures:

FIELD	DESCRIPTION
Full Name	Patient's full legal name
Date of Birth	DD/MM/YYYY format
Notes	Free-text clinical notes (allergies, comorbidities, etc.)

Tap **Save** to store the record. Tap **Edit** on the Patient Detail screen to modify an existing record.

3.3 Eye Records

Each patient can have multiple eye records (typically OD and OS). Eye records are created from the Patient Detail screen by tapping **+ Add Eye**.

Eye Record Fields

FIELD	UNIT	DESCRIPTION
Eye	—	OD (Right) or OS (Left)

FIELD	UNIT	DESCRIPTION
K1 Power	Dioptres (D)	Flat corneal meridian power
K1 Axis	Degrees (°)	TABO axis of flat meridian (0–180°)
K2 Power	D	Steep corneal meridian power
IOL Power	D	Spherical IOL power planned
IOL Cylinder	D	Toric IOL cylinder power at the IOL plane
Target Axis	°	Planned axis for IOL alignment (TABO)
SIA	D	Surgeon's expected surgically-induced astigmatism
SIA Axis	°	Axis of expected SIA
Post-Refractive	Toggle	Enable if eye had prior refractive surgery
Procedure Type	LASIK/PRK/RK	Type of prior refractive procedure

Tip

Enter K1 as the flattest meridian (lower power) and K2 as the steepest (higher power) in the standard convention. The calculator handles the vector arithmetic automatically.

CHAPTER 4

Toric Calculator

Computing optimal IOL cylinder power and alignment axis.

4.1 Entering Biometric Data

Navigate to the Toric Calculator from the Patient Detail screen. Biometric data already saved in the eye record is pre-populated automatically. You can adjust any value before calculating.

The calculator requires:

- **K1 Power & Axis** — flat keratometry value
- **K2 Power** — steep keratometry value
- **IOL Cylinder** — cylinder power of the chosen toric IOL
- **SIA** — your expected surgically-induced astigmatism magnitude
- **SIA Axis** — typical incision axis (e.g. 90° for superior incision)
- **Target Residual** — desired post-op residual astigmatism (usually 0.00 D)

4.2 PCA Correction (Barrett Posterior Corneal Astigmatism)

The **PCA Correction toggle** applies the Barrett method for posterior corneal astigmatism (PCA). This accounts for the roughly 0.3 D against-the-rule (ATR) contribution of the posterior corneal surface that is not captured by standard anterior keratometry.

> **When to enable PCA Correction**
> Enable PCA correction for all patients unless you have direct posterior corneal measurements from a Scheimpflug topographer (e.g. Pentacam). If total corneal refractive power (TCRP) data is available, enter it directly and disable the PCA toggle.

PCA Vector Calculation

```
// Double-angle vector components of anterior astigmatism
Ax = (K1 − K2) · cos(2 × K1_axis_rad)
Ay = (K1 − K2) · sin(2 × K1_axis_rad)

// Add 0.3 D ATR posterior contribution (along 0°/180° axis)
```

```
Tx = Ax + 0.3
Ty = Ay

// Convert back to magnitude/axis
adjustedMag = √(Tx² + Ty²)
adjustedAxis = ½ · atan2(Ty, Tx) × (180/π) [mod 180°]
```

The adjusted astigmatism is then used in place of the raw anterior keratometry for IOL selection, typically reducing the required cylinder power for with-the-rule (WTR) astigmatism cases.

4.3 Post-Refractive Surgery Mode

When the eye has had prior corneal refractive surgery (LASIK, PRK, or RK), standard keratometry underestimates the true corneal power. Enable the **Post-Refractive toggle** in the eye record and select the procedure type.

PROCEDURE	CONSIDERATION
LASIK / PRK	Corneal power is reduced; use adjusted axial length and history-based formulas. A warning banner is shown in the calculator prompting the use of dedicated post-refractive formulas (e.g. Barrett True-K, Haigis-L).
Radial Keratotomy (RK)	Highly variable corneal shape and diurnal fluctuation. Treat results as estimates only.

Post-Refractive Cases

Standard toric formulas are less reliable after corneal refractive surgery. Always cross-reference with a dedicated post-refractive IOL calculator and consider a conservative cylinder selection.

4.4 Results & Surgical Plan Export

After tapping **Calculate**, the results panel shows:

- **Recommended target axis** for IOL alignment
- **Expected residual astigmatism** at the corneal plane
- **Correction index estimate**
- **Rotation effect chart** — showing residual astigmatism vs. IOL rotation (0°–30°)

Tap **Export Plan** to generate a PDF surgical plan containing patient details, biometry, calculated axes, and the rotation effect chart. The PDF can be shared via any installed share sheet (email, AirDrop, etc.).

Rotation Effect Chart

The chart shows how much residual astigmatism will remain if the IOL is rotated off-axis. A steep slope indicates the chosen IOL is sensitive to rotation — consider a lower cylinder power if the patient has poor pupil dilation or a complex anatomy.

CHAPTER 5

Camera & Eye Capture

Capturing a clinical-quality anterior segment image.

5.1 Instrument Gauge Overlay

The Camera screen features a **premium instrument gauge overlay** modelled on clinical slit-lamp graticules. The overlay renders entirely in the SVG layer on top of the live camera feed.

Dial Components

ELEMENT	COLOUR	DESCRIPTION
Outer accent rings (×3)	Gold	Decorative concentric rings at R=142, 134, and 127 px
Main guide ring	Gold	Primary 120 px radius ring — align the limbus to this
Tick marks (major)	Gold	At every 30°, longer and bolder. Labelled with degree value.
Tick marks (medium)	Semi-transparent gold	At every 10°, medium length
Tick marks (minor)	Faint gold	At every 5°, short
Axis labels	Gold	TABO degrees: 0°, 30°, 60°, 90°, 120°, 150°, 180°
Inner pupil guide	Cyan dashed	54 px radius dashed ring — align the pupil margin to this
Crosshairs	White (30% opacity)	Horizontal and vertical centre guides
Centre readout	Gold / Green	Shows current tilt angle; turns green when within ±3°

5.2 Leveling the Device

The app reads the device's **roll angle** from the built-in gyroscope (DeviceMotion sensor) at 10 Hz. The centre readout displays the live tilt in degrees.

Best Practice

Hold the device horizontally and steady. The centre circle turns

green

and the shutter button glows green when the device tilt is ≤ ±3°. Capture the image only when level to ensure the TABO axis markings in the photo correspond to actual anatomical axes.

Capture Steps

1. Open the Camera screen from the Patient Detail page.
2. Position the phone over the patient's eye. Align the limbus to the main gold ring and the pupil to the cyan inner ring.
3. Wait for the centre readout to show **LEVEL** (tilt ≤ ±3°).
4. Tap the shutter button (glows green when level). The app saves the image and proceeds to the Alignment screen.

CHAPTER 6

IOL Alignment Screen

Using the alignment overlay in the operating room.

6.1 Overlay Elements

After capturing the eye photo, the Alignment Screen superimposes a surgical overlay directly on the image. This screen is designed to be used at the operating microscope — place the phone where both the overlay and the patient's eye are visible simultaneously.

OVERLAY ELEMENT	COLOUR	MEANING
Reference axis line	Blue	The pre-operative corneal reference meridian (K1 axis)
Target axis line	Green	The desired final IOL axis — align the IOL markings to this
Current axis line	Gold	The current device/manual rotation angle as adjusted
Limbal ring	Cyan	Guide circle for limbal registration
Gold tick marks	Gold	Degree markers every 10° around the limbal ring
Landmark dots	Colour-coded	Pre-marked vessel reference points (if annotations saved)

6.2 Manual Axis Adjustment

Use the **+/–** degree buttons to manually rotate the overlay axes in 1° increments. This allows fine-tuning if the captured image is slightly rotated or if you need to adjust for cyclorotation during surgery.

6.3 Landmark Annotations

If vessel landmarks were annotated before capture (see Chapter 7), they appear as coloured ring-and-dot markers on the alignment overlay. These help confirm the correct rotational orientation of the eye against the pre-operative baseline.

Quick Actions

- **Recapture** — return to Camera to take a new photo without losing the patient record.

- **Annotate** — open the Landmark Annotation screen to add or edit vessel markers.

CHAPTER 7

Landmark Annotation Tool

Marking vessel reference points for intraoperative orientation.

7.1 Overview

Cyclorotation of the eye during surgery — typically 2°–5° — can shift the effective axis placement of a toric IOL. Marking pre-operative conjunctival vessel landmarks provides a reliable reference to compensate for this rotation intraoperatively.

7.2 Adding Landmarks

1. Navigate to **Landmark Annotation** from the Patient Detail screen (Annotate button) or from the Alignment Screen quick actions.
2. The captured eye photo is shown full-screen. **Tap any vessel** or anatomical landmark on the image to place a marker.
3. Each marker is assigned a sequential number and a unique colour. Up to 6 colours cycle automatically.
4. Use **Undo Last** to remove the most recent marker, or **Clear All** to start over.
5. Tap **Save** to store the landmarks. They will appear in all future Alignment overlays for this eye.

> **Clinical Tip**
> Mark 3–4 distinct conjunctival vessels at different positions around the limbus (e.g. 3 o'clock, 6 o'clock, 9 o'clock). Avoid marking the limbus itself — vessels are more reliable as they are fixed relative to the underlying sclera.

7.3 Landmark List

Below the image, a list shows all placed landmarks with their:

- Sequential number and colour indicator
- Normalised coordinates (x%, y%) as a percentage of the image width/height

These coordinates are stored in the patient record and are resolution-independent.

CHAPTER 8

Alpins Vector Analysis

Quantifying post-operative astigmatism outcomes.

8.1 What is Alpins Analysis?

The **Alpins method** (developed by Prof. Noel Alpins) is the internationally recognised standard for reporting astigmatism surgery outcomes. It uses double-angle vector mathematics to quantify the relationship between what was planned (TIA), what was achieved (SIA), and what remains (DV).

Access Alpins Analysis from the Patient Detail screen using the **Alpins** button. The TIA fields are pre-filled from the eye's corneal data (with PCA correction if applicable).

Pre-operative vs Post-operative use

For real post-operative analysis, enter the patient's post-op refraction cylinder as the

SIA

values. The pre-filled SIA (from IOL cylinder) gives a theoretical analysis assuming perfect placement — useful for planning, but not for true outcome assessment.

8.2 Metrics Reference

TIA — Target-Induced Astigmatism

The astigmatic change the surgery was intended to induce. Derived from pre-op corneal astigmatism and IOL plan.

SIA — Surgically-Induced Astigmatism

The astigmatic change that was actually induced, measured from post-op refraction.

DV — Difference Vector

The residual astigmatism error. Ideal value: 0.00 D. This is the vector from TIA endpoint to SIA endpoint.

METRIC	IDEAL VALUE	GREEN	YELLOW	RED
CI — Correction Index	1.00	0.9 – 1.1	0.75 – 1.25	< 0.75 or > 1.25
ME — Magnitude Error	0.00 D	≤ ±0.25 D	> ±0.25 D	—

METRIC	IDEAL VALUE	GREEN	YELLOW	RED
AE — Angle Error	0.0°	≤ ±5°	> ±5°	—
IS — Index of Success	0.00	< 0.2	0.2 – 0.5	> 0.5

Interpretation Guide

CI VALUE	INTERPRETATION	IMPLICATION
> 1.0	Overcorrection	SIA exceeded TIA — IOL cylinder power was too strong or axis was over-rotated
= 1.0	Perfect correction	Exact target achieved
< 1.0	Undercorrection	SIA less than TIA — consider higher cylinder next time or check for axis error

8.3 Double-Angle Plot

The polar plot below the metrics shows all three vectors (TIA, SIA, DV) in double-angle space (each axis is doubled to convert the 0°–180° TABO range to a full 0°–360° circle). Key features:

- 🟦 **Blue dashed line** — TIA vector
- 🟩 **Green dashed line** — SIA vector
- 🟥 **Red solid line** — DV (from TIA tip to SIA tip)
- Concentric grid circles at 33%, 67%, and 100% of the maximum vector magnitude
- A short DV indicates a good outcome; a long DV indicates a residual error.

CHAPTER 9

SIA Nomogram Tracker

Building your personalised surgically-induced astigmatism profile.

9.1 Overview

Every surgeon induces a slightly different amount of astigmatism through their incision — influenced by incision size, location, technique, and individual healing response. The **SIA Nomogram** tracks your personal achieved SIA across cases and calculates your **vector mean**, allowing you to personalise future calculations.

Access the SIA Nomogram from the Patient List header (gold **SIA** button).

9.2 Adding a Case

1. Tap **+ Add Case**.
2. Enter the **Planned SIA** (magnitude and axis) — what you expected to induce.
3. Enter the **Achieved SIA** (from post-op refraction) — what actually occurred.
4. Add optional notes (incision type, wound size, etc.).
5. Tap **Save**. The mean is recalculated instantly.

9.3 Mean SIA Display

The top of the screen shows your personal **Vector Mean SIA**:

- **Mean magnitude** — average astigmatism induced (in Dioptres)
- **Mean axis** — predominant direction of your induced astigmatism
- **Case count** — number of cases contributing to the average

Each case in the list is colour-coded by the difference between planned and achieved SIA:

Green

= within 0.25 D,

Yellow

= within 0.50 D,

Red

= > 0.50 D difference.

Clinical Recommendation

Build your nomogram with at least

20 cases

before using the mean SIA value in routine calculations. Early cases may show higher variance as your technique standardises. Review the tracker periodically and consider separate profiles for different incision locations (e.g. temporal vs. superior).

CHAPTER 10

Settings

Configuring the application to your preferences.

10.1 Accessing Settings

Tap the ⚙ gear icon in the top-right of the Patient List screen.

10.2 Available Settings

SETTING	OPTIONS	DESCRIPTION
Default SIA	0.00 – 1.00 D	Pre-populates the SIA field in new eye records
Default SIA Axis	0° – 180°	Pre-populates the SIA axis for your typical incision location
PCA Correction	On / Off	Global default for the PCA toggle in the Toric Calculator
Surgeon Name	Free text	Appears on exported PDF surgical plans
Institution	Free text	Appears on exported PDF surgical plans

CHAPTER 11

Clinical Reference

Quick-reference formulae and axis conventions.

11.1 TABO Axis Convention

All axes in the Zahran Toric Tool follow the **TABO convention**:

DEGREES	POSITION ON CLOCK FACE	ANATOMICAL DIRECTION
0° / 180°	3 o'clock / 9 o'clock	Horizontal
45°	1:30 o'clock	Oblique (up-right)
90°	12 o'clock	Vertical (superior)
135°	10:30 o'clock	Oblique (up-left)

With-the-Rule (WTR) astigmatism has its steep meridian near 90° (vertical).
Against-the-Rule (ATR) astigmatism has its steep meridian near 180°/0° (horizontal).

11.2 Double-Angle Vector Method (DAVM)

All vector operations in the app use the DAVM to allow standard trigonometry on the 0°–180° TABO range:

```
J0 = −(C/2) · cos(2 × axis_rad) // WTR/ATR Jackson cross-cylinder
J45 = −(C/2) · sin(2 × axis_rad) // Oblique component

// Vector sum of two astigmatisms A and B:
Rx = A.C·cos(2·A.axis) + B.C·cos(2·B.axis)
Ry = A.C·sin(2·A.axis) + B.C·sin(2·B.axis)
Result.C = √(Rx² + Ry²)
Result.axis = ½ · atan2(Ry, Rx) · (180/π) [mod 180°]
```

11.3 Recommended Data Entry Ranges

PARAMETER	TYPICAL RANGE	ALERT RANGE
Corneal astigmatism (K1–K2)	0.25 – 3.00 D	> 5.00 D — check for keratoconus
IOL cylinder power	1.00 – 6.00 D	> 6.00 D — rarely available
SIA	0.10 – 0.80 D	> 1.00 D — verify incision size/type
Target axis	0° – 180°	—

CHAPTER 12

Troubleshooting

Common issues and their solutions.

12.1 Common Issues

PROBLEM	CAUSE	SOLUTION
Camera screen is black	Camera permission not granted	Go to device Settings → Privacy → Camera → enable for Zahran Toric Tool
Tilt indicator not responding	Motion sensor unavailable	Ensure the app has motion access. Restart the app if needed.
Calculator result seems incorrect	K1 entered as steeper than K2	K1 must be the flat meridian (lower dioptres). Swap K1/K2 values.
Alpins CI shows extreme value (>2.0)	SIA fields left as IOL defaults	Enter actual post-operative refraction cylinder for a valid analysis.
Landmark markers missing on alignment screen	Landmarks not saved	In Landmark Annotation, verify you tapped Save before closing the screen.
PDF export fails	No share sheet available	Ensure Files or another share target is installed. Try again after restarting the app.
PCA toggle not visible	K values not entered in eye record	PCA correction requires K1 Power, K1 Axis, and K2 Power to be filled in the eye record.
Patient data not persisting	Storage permission issue	All data is stored on-device. If data is missing after app restart, check available storage space.

12.2 Data & Privacy

All patient data is stored **locally on your device only**. No data is transmitted to any server or third party. The app does not require an internet connection to function. Uninstalling the app will permanently delete all stored patient data — ensure records are exported before uninstalling.

Data Backup

> The Zahran Toric Tool does not currently include a cloud backup feature. Export surgical plans as PDFs for important cases to maintain a record outside the app.

12.3 Contact & Support

For questions, feature requests, or bug reports, contact the development team through the app store listing or your institution's IT support channel.

Zahran Toric Tool — Version 1.0.0

For clinical decision support only. Not a substitute for professional medical judgment.